
DATA SCIENCE PROJECT

PREDICTING REAL ESTATE LISTING SUCCESS AND PRICE OPTIMIZATION USING PROPERTY DESCRIPTIONS

NAME: REDDY ABINAYA

STUDENT ID: 7235SYXD

GITHUB LINK: <https://github.com/7235SYXD/Real-Estate.git>





PROJECT OVERVIEW

Background research questions and project objectives

PROJECT BACKGROUND & RESESRCH QUESTION

Background

- The US residential real estate market generates millions of listings annually
- This project predicts price and status by fusing structured attributes with NLP signals from descriptions
- Four Kaggle datasets: SAKIB and POLARTECH (structured); TEXAS and NEW YORK (descriptions)
- Modeling limited to TX and NY — full corpus rows have descriptions

Research Question

- "Which structural and linguistic features most strongly predict sale price and listing success in TEXAS and NEW YORK, and how can SHAP insights generate actionable recommendations?"

Scope

- Two tasks: sale price (regression) and listing success (classification)
 - Focus on structural and NLP features in Texas and New York markets
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PROJECT OBJECTIVES

- To determine the VADER sentiment polarity, Flesch-Kincaid readability scores, luxury keyword count, YAKE key phrases carry a measurable association with listing price and listing success outcome in the market.
- Evaluate if the Optuna fine-tuning improves the overall accuracy over the default baseline, and which model performs the best.
- Assess whether the stacking ensemble outperforms the base models on both tasks, which can quantify the model's contribution through Ridge meta-learner coefficient analysis.
- To quantify how NLP features add prediction accuracy beyond the structural attributes alone.
- Analyze which structural and NLP features drive a strong predicted price and listing success using SHAP TreeExplainer for tree models and permutation importance for the MLP neural network.





PROJECT PLAN

7-WEEK SCHEDULE



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TASK	ACTIVITIES	DATES
Setup & data Ingestion Literature Review DataCleaning	Google Colab setup, upload all the 4 datasets from kagglehub, raw CSV audit. Reading papers for literature review.	25 May – 1 June
TX+NY Extraction NLP Feature Extraction Assessment 1 Preparation	Extract TX+NY rows only, and stratified sample to 200-300K. Apply features like Vander/Flesh/Luxury/Yake	2 June – 8 June
EDA & Visualization Baseline Models Assessment 2 preparation	Atleast 3-4 visualization after merging,cleaning,extracting the data from the 4 datasets. Baseline models like CatBoost, RandomForest, MLP	9 June – 15 June
Fine-tuned Models OOF +Stacking	Fine tune all the models and record R^2 . 5-fold OOF predictions(regression), train Record Ridge meta-learner coefficients and compare individual vs stacked R^2	16 June – 22 June
Ablation Study	Model trained on structural-only features vs structural+NLP features.	23 June – 29 June
SHAP & Interpretability	SHAP TreeExplainer on tuned model, report top-15 features per model.	30 June – 6 July
Report & Assessment 3 submission	Abstract and conclusions, incorporate feedback, Harvard references, GitHub tag. Final submission.	6 July – 10 July



DATA MANAGEMENT PLAN

DATASET OVERVIEW, COLLECTION, STORAGE, AND ETHICS



DATASET OVERVIEW & METADATA

Dataset 1: SAKIB — 2.2M rows x 10 cols, all 50 states

- Columns: price, bed, bath, house_size, acre_lot, state, status
- No descriptions — structural backbone

Dataset 2: POLARTECH — ~500K rows x 28 cols, all 50 states

- Columns: price, bedrooms, bathrooms, sqft, state, city
- No descriptions — structural supplement

Dataset 3: TEXAS 2026 — ~12K rows x 13 cols, TX only

- Columns: description, price, sqft, bedrooms, bathrooms, home_type
- Real descriptions — NLP source for Texas

Dataset 4: NEW YORK 2026 — ~9K rows x 11 cols, NY only

- Columns: description, price, sqft, beds, baths, status
- Real descriptions — NLP source for New York

Metadata

- Formats: CSV data, .py/ipynb code, .pdf report; storage ~600MB-1GB
 - No PII in any dataset
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VERSION CONTROL, SECURITY & ETHICS

Version Control

- GitHub: <https://github.com/7235SYXD/Real-Estate.git>
- One commit per week, snake_case naming, version tags (v1.0, v2.0)
- README.md in Week 7 with overview and setup instructions

Security & Storage

- Primary: Public GitHub; Secondary: OneDrive for backup
- Datasets via KaggleHub links

Ethics & Compliance

- No PII; all data publicly available on Kaggle for academic use
 - Geographic bias analyzed and transparently reported
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REFERENCES

Dorogush, A.V., Ershov, V. and Gulin, A. (2018) 'CatBoost: Gradient boosting with categorical features support', Workshop on ML Systems at NeurIPS 2017. Available at: <https://arxiv.org/abs/1810.11363> (Accessed: 27 May 2026).

Hutto, C.J. and Gilbert, E. (2014) 'VADER: A parsimonious rule-based model for sentiment analysis of social media text', Proceedings of the 8th AAAI Conference on Weblogs and Social Media, pp. 216-225. Available at: <https://ojs.aaai.org/index.php/ICWSM/article/view/14550> (Accessed: 29 May 2026).

Zhao, H. and Wang, K. (2023) 'Predicting real estate price using stacking-based ensemble learning', American Journal of Information Science and Technology, 7(2), pp. 70–75. Available at: <https://doi.org/10.11648/j.ajist.20230702.14> (Accessed: 2 June 2026).



**Thank
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